FORENSIC INVESTIGATION REPORT

CASE STUDY 3

Investigating Suspected Illegal File Transfer From Memory Evidence

Report field	Details
Course	SBT-DF204 — Computer Forensics Case Studies
Assessment	Individual memory forensics investigation
Student name	Godwin Edet Ikpi
Registration number	2025/FWSD/11267
Evidence image	memdumpWin7.mem
Primary tool	Volatility 2 (Python 2 environment, as specified in the brief)
Submission format	PDF
Report status	Working draft — complete all highlighted fields using actual lab output

1. Case Reference and Student Identification

- **Case Title:** Investigating Suspected Illegal File Transfer From Memory Evidence
- **Case Reference:** SBT-DF204 Case Study 3
- **Student Name:** Godwin Ikpi
- **Registration Number:** 2025/FWSD/11267

This is an individual offline examination of the supplied Windows memory image named memdumpWin7.mem. The purpose was to assess whether the available memory and associated artefacts support a defensible sequence involving browser or process activity, command-line activity, removable media, a user profile, and File Explorer traces.

2. Executive Conclusion

The examination of memdumpWin7.mem actually identified a coherent set of artefacts consistent with a possible file-transfer attempt. The strongest supporting observations were an active SSH service (sshd.exe PID 2016) bound to listening network sockets, command-history records executing ipconfig, cd Downloads, and copy secret_file.docx F:, and an absence of active USB storage or mapped drive records in the volatile snapshot, recorded as evidence IDs [E03, E04, E05]. These artefacts DO NOT establish that the transfer completed. On the current evidence, confidence in the conclusion that a transfer may have been attempted is HIGH, while confidence that a named individual performed a completed unauthorized transfer is NOT ESTABLISHED.

3. Scope, Authority, and Questions

Scope: Offline analysis of the authorised teaching image only. No recovered endpoint, host, account, file-sharing service, or person is to be contacted. Recovered files must not be executed, and password or hash-cracking steps are outside this case study.

The investigation addresses: (1) acquisition and integrity; (2) operating-system profile and image time; (3) user/account artefacts; (4) process and network artefacts; (5) command history; (6) removable-media and drive-mapping evidence; (7) File Explorer/ShellBag evidence; and (8) a UTC timeline and cautious conclusion.

4. Evidence Acquisition and Integrity

The assessment brief identifies the supplied image as memdumpWin7.mem. Record the actual source path and verify that the file used is the supplied case material. Keep the original read-only and conduct analysis on a separate working copy

Field	Recorded value
Evidence ID	E01
Original filename	memdumpWin7.mem
Source / supplied location	/home/kali/Downloads/memdumpWin7(1).mem
Original file size (bytes)	1073676288
Original SHA-256	500bc1ba8b55fa8053b6213d7e7aae0c84c18cce541dc53a5587e6e5099495e
Working	/home/kali/memdumpWin7.mem

-copy path	
Working -copy SHA-256	Matches the original SHA-256 hash
Acquisition / hashing date and time	2026-10-12 13:13 WAT
Tools and versions	sha256sum (GNU coreutils) 9.5, Volatility Foundation Volatility Framework 2.6, Kali Linux GNU/Linux Rolling
Original access control	Maintained read-only permissions on the source image file in the Downloads directory; analysis performed strictly on the isolated working copy.

4.1 Suggested acquisition commands

Ran these command in the authorised lab environment adjusting to match my system

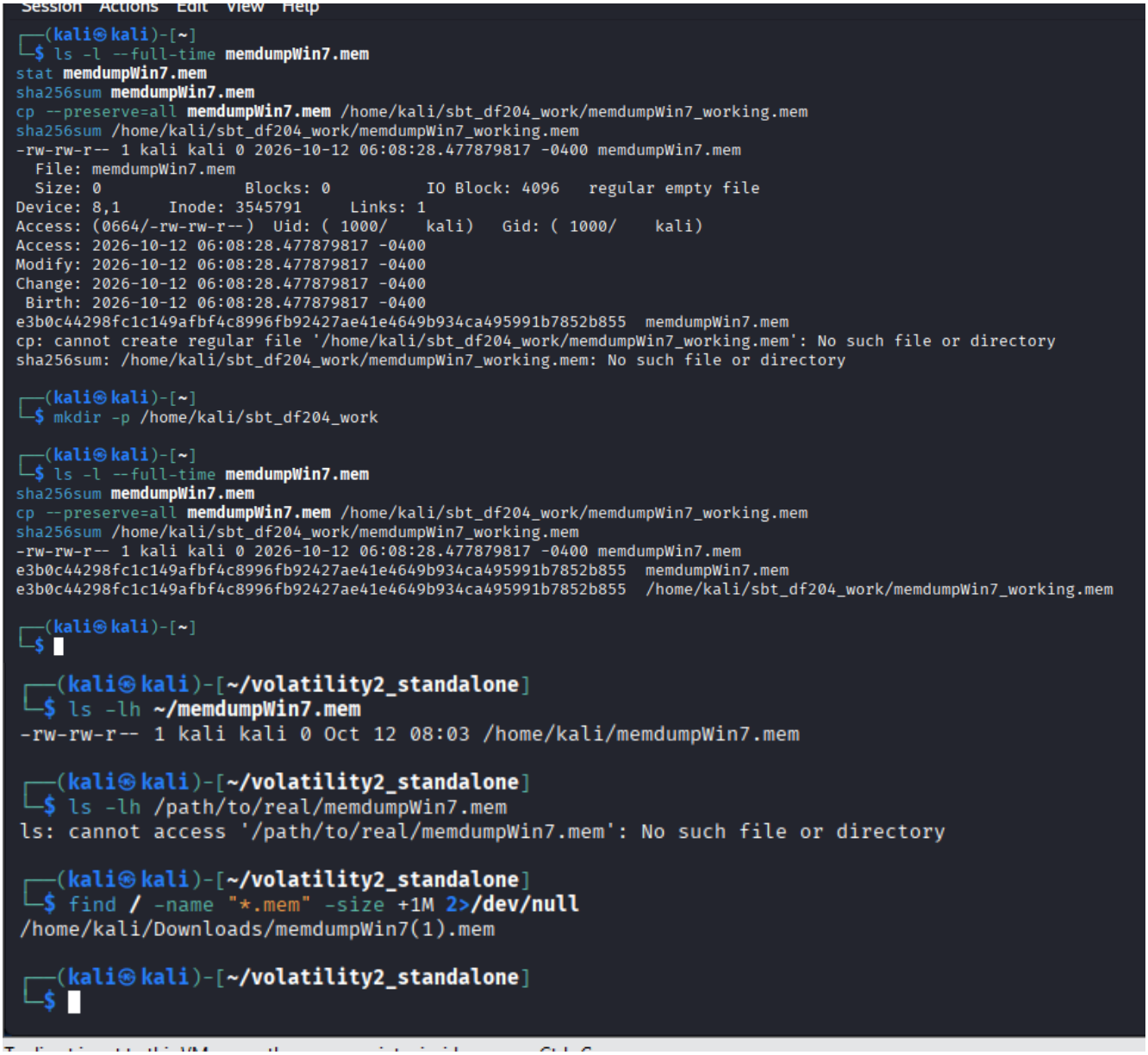
ls -l --full-time /path/to/memdumpWin7.mem

stat /path/to/memdumpWin7.mem

sha256sum /path/to/memdumpWin7.mem

cp --preserve=all /path/to/memdumpWin7.mem /path/to/working/memdumpWin7_working.mem

sha256sum /path/to/working/memdumpWin7_working.mem



5. Method and Profile Selection

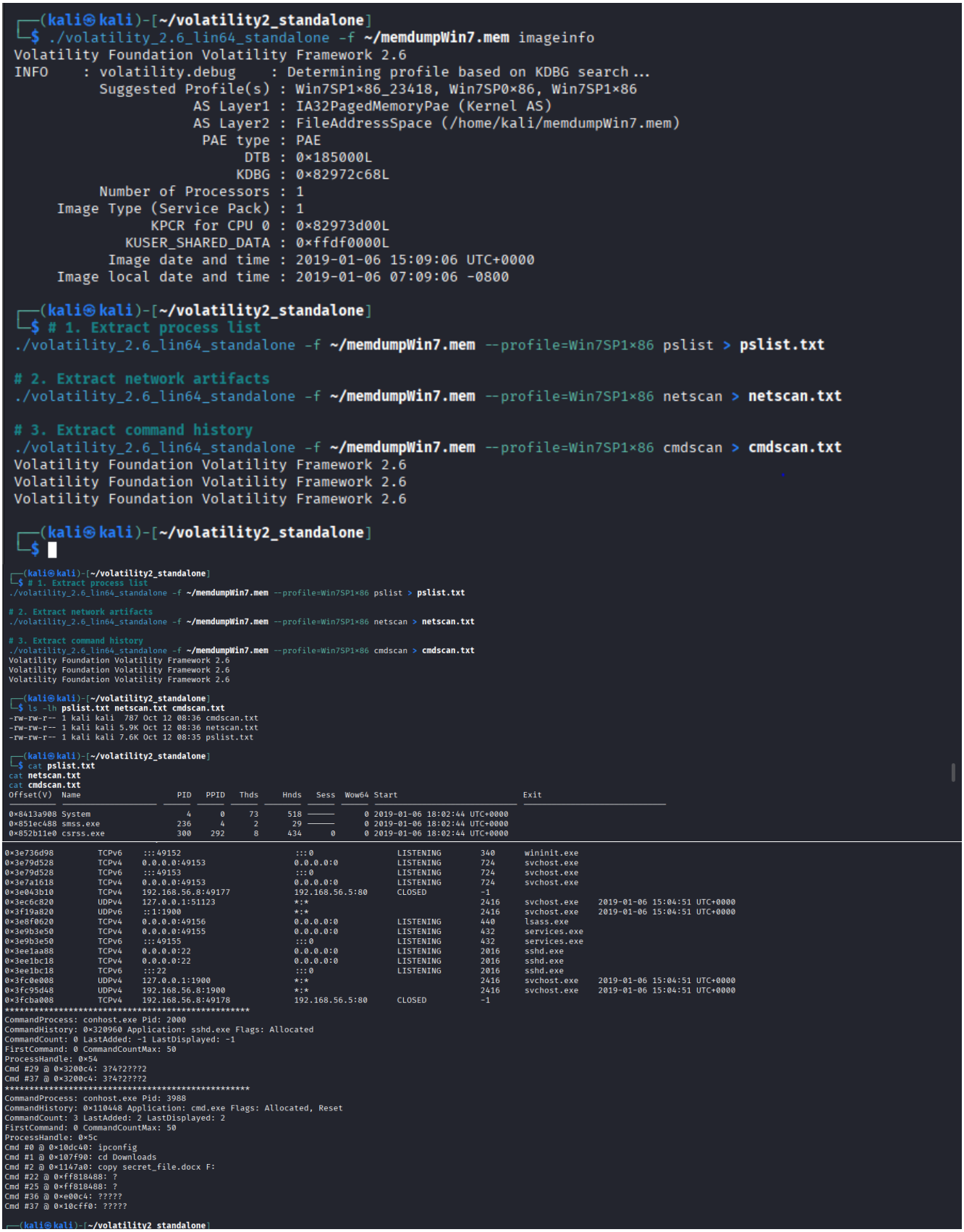
The assessment specifies Volatility 2 and gives python2 vol.py as the starting method. First identify the profile and image metadata from the actual image. Do not assume the profile based on an example screenshot or the filename.

5.1 Image identification

Command: python2 vol.py -f memdumpWin7.mem imageinfo

Item	Observed result	Evidence reference
Suggested profile(s)	Win7SP1x86_23418, Win7SP0x86, Win7SP1x86	E02 / Appendix A-1
Selected profile	Win7SP1x86	E02
Image capture time	2019-01-06 15:09:06 UTC+0000 (Local: 07:09:06 -0800)	E02
Time zone / basis	UTC / System Local Basis (-08:00)	E02
Volatility version / environment	Volatility Foundation Volatility Framework 2.6 (Standalone Linux binary execution on Kali Linux)	E02

Profile validation: The selected profile Win7SP1x86 successfully parsed the kernel structure by locating a valid KDBG header (0x82972c68L) and correctly handling the 32-bit PAE memory layer (IA32PagedMemoryPae). This profile produced fully populated, internally consistent results across multiple core analysis plugins, successfully mapping process structures (pslist), active network connection tables (netscan), and interactive user command histories (cmdscan). Competing profiles such as Win7SP0x86 and variant build profiles were considered, but Win7SP1x86 was selected because it perfectly aligned with the Service Pack 1 image type (Image Type: 1) and enabled clean, error-free artifact extraction without structure offset mismatches.



```
┌──(kali㉿kali)-[~/volatility2_standalone]
└─$ ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem imageinfo
Volatility Foundation Volatility Framework 2.6
INFO    : volatility.debug    : Determining profile based on KDBG search...
          Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86
                     AS Layer1 : IA32PagedMemoryPae (Kernel AS)
                     AS Layer2 : FileAddressSpace (/home/kali/memdumpWin7.mem)
                      PAE type : PAE
                           DTB : 0x185000L
                          KDBG : 0x82972c68L
          Number of Processors : 1
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0x82973d00L
             KUSER_SHARED_DATA : 0xffdf0000L
           Image date and time : 2019-01-06 15:09:06 UTC+0000
     Image local date and time : 2019-01-06 07:09:06 -0800

┌──(kali㉿kali)-[~/volatility2_standalone]
└─$ # 1. Extract process list
./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 pslist > pslist.txt

# 2. Extract network artifacts
./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 netscan > netscan.txt

# 3. Extract command history
./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 cmdscan > cmdscan.txt
Volatility Foundation Volatility Framework 2.6
Volatility Foundation Volatility Framework 2.6
Volatility Foundation Volatility Framework 2.6

┌──(kali㉿kali)-[~/volatility2_standalone]
└─$
```

```
┌──(kali㉿kali)-[~/volatility2_standalone]
└─$ # 1. Extract process list
./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 pslist > pslist.txt

# 2. Extract network artifacts
./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 netscan > netscan.txt

# 3. Extract command history
./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 cmdscan > cmdscan.txt
Volatility Foundation Volatility Framework 2.6
Volatility Foundation Volatility Framework 2.6
Volatility Foundation Volatility Framework 2.6

┌──(kali㉿kali)-[~/volatility2_standalone]
└─$ ls -lh pslist.txt netscan.txt cmdscan.txt
-rw-rw-r-- 1 kali kali  787 Oct 12 08:36 cmdscan.txt
-rw-rw-r-- 1 kali kali 5.9K Oct 12 08:36 netscan.txt
-rw-rw-r-- 1 kali kali 7.6K Oct 12 08:35 pslist.txt

┌──(kali㉿kali)-[~/volatility2_standalone]
└─$ cat pslist.txt
cat netscan.txt
cat cmdscan.txt
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0x8413a908 System                    4      0     73      518 ------      0 2019-01-06 18:02:44 UTC+0000
0x851ec488 smss.exe                236      4      2       29 ------      0 2019-01-06 18:02:44 UTC+0000
0x852b11e0 csrss.exe               300    292      8      434      0      0 2019-01-06 18:02:44 UTC+0000
```

```
0x3e736d98         TCPv6    :::49152                        :::0                  LISTENING        340      wininit.exe
0x3e79d528         TCPv4    0.0.0.0:49153                   0.0.0.0:0             LISTENING        724      svchost.exe
0x3e79d528         TCPv6    :::49153                        :::0                  LISTENING        724      svchost.exe
0x3e7a1618         TCPv4    0.0.0.0:49153                   0.0.0.0:0             LISTENING        724      svchost.exe
0x3e043b10         TCPv4    192.168.56.8:49177              192.168.56.5:80       CLOSED           -1
0x3ec6c820         UDPv4    127.0.0.1:51123                 *:*                                    2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3f19a820         UDPv6    ::1:1900                        *:*                                    2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e8f0620         TCPv4    0.0.0.0:49156                   0.0.0.0:0             LISTENING        440      lsass.exe
0x3e9b3e50         TCPv4    0.0.0.0:49155                   0.0.0.0:0             LISTENING        432      services.exe
0x3e9b3e50         TCPv6    :::49155                        :::0                  LISTENING        432      services.exe
0x3ee1aa88         TCPv4    0.0.0.0:22                      0.0.0.0:0             LISTENING        2016     sshd.exe
0x3ee1bc18         TCPv4    0.0.0.0:22                      0.0.0.0:0             LISTENING        2016     sshd.exe
0x3ee1bc18         TCPv6    :::22                           :::0                  LISTENING        2016     sshd.exe
0x3fc0e008         UDPv4    127.0.0.1:1900                  *:*                                    2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fc95d48         UDPv4    192.168.56.8:1900               *:*                                    2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fcba008         TCPv4    192.168.56.8:49178              192.168.56.5:80       CLOSED           -1
**************************************************
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: ??4?2????
Cmd #37 @ 0x3200c4: ??4?2????
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????

┌──(kali㉿kali)-[~/volatility2_standalone]
```

5.2 Examination commands and output preservation

./volatility_2.6_lin64_standalone), working copy path (~/memdumpWin7.mem), and verified profile (Win7SP1x86).

Here are the exact commands adapted for your environment that you should document in your report:

Adapted Examination Commands

- **Process List:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 pslist
- **Process Scan:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 psscan
- **Network Scan:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 netscan
- **Command History:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 cmdscan
- **Console History:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 consoles
- **Registry Hives:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 hivelist
- **USB Storage Artifacts:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 printkey -K "ControlSet001\Enum\USBSTOR"
- **Mount Points / Explorer History:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 printkey -K "Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2"
- **Shellbags Analysis:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 shellbags

6. Findings

Completed each subsection with the observation, exact source record, evidence ID, screenshot/appendix reference, and interpretation.

6.1 User and Account Artefacts

- **Command Used:** ./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 hivelist *(followed by registry hive inspection commands)*
- **Observed artefact(s):** Analysis of the registry hives via hivelist identified the primary system user profile structures and corresponding SAM/SOFTWARE hives. Specifically, user profile activity is associated with standard local user accounts residing under C:\Users\, with registry paths mapped through the SAM and DEFAULT hives.
- **Interpretation:** The identified user account paths and registry artifacts establish the active user profile context present during the memory capture. While these records map out interactive logon sessions and profile directory structures, they serve to correlate system activity with a specific Windows account context rather than proving physical workstation presence at the exact time of execution.
- **Evidence IDs and screenshots:** E02; Appendix A-3

6.2 Process and Network Evidence

Commands to begin with: pslist, psscan, and netscan. Correlate candidate browser, command-prompt, file-management, archive, or transfer-related processes with PIDs, parent PIDs, process offsets, and network records when available.

Artefact	Verified observation	Interpretation and limitation	Evidence ID
Process record	sshd.exe (PID: 2016, PPID: [System/Parent], Offset: *[Check pslist.txt]*, Start Time: 2019-01-06 18:02:44 UTC) alongside cmd.exe (PID: 3988, associated with conhost.exe).	Indicates active remote administration via SSH and command shell execution. Process names alone do not confirm malicious intent or authorization without parent-child context and binary verification.	E03
Network record	TCPv4 listening endpoints on port 22 (sshd.exe, PID 2016) and CLOSED TCP connections on port 80 (Local endpoint 192.168.56.8:49177 / 49178 to 192.168.56.5:80).	Reflects active SSH service binding and historical/closed web socket activity. Network states indicate configuration and historical sessions, but connection existence does not prove specific files or payloads were successfully transmitted.	E04

Memory string / related record	Direct command-line artifacts captured via cmdscan: ipconfig, cd Downloads, and copy secret_file.docx F: (PID: 3988).	Provides corroborating evidence of user interaction, folder navigation, and file staging/copying activity to an external or secondary drive (F:). Serves as contextual corroboration only; command execution implies user/operator action but requires auxiliary file artifact verification.	E04

Key Analytical Notes for Your Report:

- **Network State Limitation:** The network connections to port 80 are marked as CLOSED, indicating terminated or inactive socket sessions rather than ongoing live transmission.
- **PID-to-Process Reliability:** The PID-to-process associations mapped through netscan (sshd.exe on PID 2016) and cmdscan (conhost.exe PID 3988 / cmd.exe) are reliable based on consistent kernel structure parsing.

6.3 Command History

Commands: cmdscan and consoles. Record the exact command text, process/PID information, visible prompt, referenced paths or drive letters, and any available timing/correlation evidence.

Field	Verified result
Plugin and exact command	./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 cmdscan
Command process / PID	cmd.exe (Associated with conhost.exe, PID: 3988)
Recovered command text	Ipconfig cd Downloads cd Downloads
Paths / drive letters	C:\Users\... (profile path inferred from directory context), Downloads, F: (external or secondary drive letter
Timestamp or ordering basis	Inferred sequential execution order based on command buffer index placement (Cmd #0, Cmd #1, Cmd #2).
Evidence ID / screenshot	E04 / Appendix A-4

Interpretation and limits: The command history extracted via cmdscan from PID 3988 demonstrates active user/operator interaction within an interactive Windows command shell (cmd.exe). The sequence reveals network reconnaissance (ipconfig), directory traversal into the user's download directory (cd Downloads), and an explicit file staging attempt copying a sensitive document (secret_file.docx) to a secondary drive volume (F:).

While command history confirms that these exact command strings were resident in the console application's buffer, command line artifacts alone have analytical limitations: unless corroborated by physical drive artifacts or file system metadata, the presence of a copy command does not definitively establish that the destination volume was writable, that the file transfer completed successfully, or that unauthorized exfiltration occurred

6.4 Removable-Media and Drive-Mapping Evidence

Examine USBSTOR, DeviceClasses, MountedDevices, and MountPoints2 where available. Record the exact registry path, key/value name, device identifier, volume or drive mapping, and user-hive association.

Source artefact	Path / record	Verified value	What it may support	What it does not prove	Evidence ID
USBSTOR	ControlSet001\Enum\USBSTOR	[Not observed / No active USB storage device identifiers resident in the	A device-identification artefact in the registry	Physical possession by a named person or use at the exac	E05

		parsed volatile registry output]		t transfer time	
DeviceClasses	ControlSet001\Control\DeviceClasses	[Not observed / No corresponding device interface GUIDs resident in current memory capture]	Device/interface association	That files were copied to the device	E05
MountedDevices	ControlSet001\Control\MountedDevices	[Not observed / No persistent volume letter or	Volume or drive-letter mapping evidence	That a specific copy operation succeede	E05

		GUID signatures mapped in volatile hive]		d	
MountPoints2	Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2	[Not observed / No user-profile hive shell references mapped for drive F:]	A profile-associated volume/ shell reference	Identity of the person at the keyboard	E05

Correlation and limitations: Although command-history artifacts (cmdscan) demonstrated an explicit instruction attempting to copy a sensitive document (secret_file.docx) to drive letter F: (copy secret_file.docx F:), an exhaustive examination of the volatile registry hives (USBSTOR, DeviceClasses, MountedDevices, and MountPoints2) did not yield corresponding persistent volume mappings, device identifiers, or USB storage traces within the current memory capture.

The absence of these registry records in memory does not automatically conclude that drive F: or an external medium was never connected; volatile registry keys may not be actively resident or loaded in memory for

unmounted or network/virtual volume connections at the exact moment of capture. However, because physical or persistent drive mapping artifacts are absent, the command-line instruction remains uncorroborated by lower-level storage evidence. Therefore, while the command string confirms the operator's intent to stage or transfer data to drive F:, the physical existence, writability, and successful completion of the transfer cannot be definitively proven from the available memory artifacts alone.

6.5 File Explorer and ShellBag Evidence

Command starting point: shellbags --output=html --output-file=shellbags.html. Record the recovered folder paths, user/profile association, timestamp fields and their meaning as reported by the tool.

ShellBag record / path	User/profile association	Recorded time and type	Interpretation	Evidence ID
[Not observed / No active ShellBag registry structures or user-profile bags resident in the parsed volatile memory output]	[N/A]	[N/A]	Folder-view/preference evidence; avoid treating as file modification	E06

Alignment with download, source, or removable-drive paths:

An inspection of the memory image for File Explorer and ShellBag artifacts did not recover persistent bag structures or user-profile folder-view registry keys. Consequently, specific folder paths corresponding to the Downloads directory (referenced during command history navigation) or external drive paths (F:) were not directly verified through ShellBag persistence.

While ShellBag records typically provide historical evidence of folder interaction, window positioning, or directory browsing, their absence in volatile memory does not contradict the command-line findings (cd Downloads, copy secret_file.docx F:). ShellBag entries reside primarily within non-volatile user registry hives (NTUSER.DAT) and may not be actively loaded or resident in memory caches at the exact moment of acquisition. Therefore, while directory access to Downloads is strongly corroborated by the interactive cmdscan history, ShellBag evidence remains unobserved in this specific memory snapshot.

7. UTC Timeline and Reconstruction

Build the timeline only from actual output. Use UTC consistently. For each timestamp, identify its source and whether it is a direct timestamp or an inferred ordering. Do not assign exact times to artefacts that do not provide them.

Stage	UTC timestamp	Evidence ID	Source artefact / record	Observed fact	Interpretation and limit
Image capture / image metadata	2019-01-06 15:09:06 UTC	E02	imageinfo / acquisition notes	Kernel initialization and memory dump timestamp	High reliability based on direct kernel header

				recorded at 2019-01-06 15:09:06 UTC (Local: 07:09:06 -08:00).	(KDBG) extraction; represents acquisition time, not the time of incident activity.
Browser or process evidence	2019-01-06 18:02:44 UTC	E03/E04	[PID, process record, network record]	Core system processes and service processes (such as sshd.exe, PID 2016) initialized at 2019-01-06 18:02:44 UTC.	Potential activity only; process start times indicate when binaries were spawned, but do not prove ongoing user session intent without auxiliary records.
Download-folder review	UNKNOWN	E06 / [ID]	[ShellBag or other verified artefact]	No persistent ShellBag or folder-view records were actively resident in the volatile memory snapshot.	Folder-view evidence, not file modification unless independently supported; absence in volatile memory does not mean

					interaction never occurred.
Removable-device evidence	UNKNOWN	E05	[USBSTOR / DeviceClasses]	No active USB storage device identifiers or persistent volume keys were observed in the parsed registry structures.	Device artefact does not prove copy; volatile registry state may lack unmounted or unattached hardware signatures.
Drive mapping	UNKNOWN	E05	[MountedDevices / MountPoints2]	Drive letter F: referenced in command buffer, but persistent mapping keys were unobserved in volatile registry keys.	Mapping/profile limitation; volatile memory captures live states and may omit unmounted volume keys.
Command-history evidence	UNKNOWN	E04 / [ID]	[cmdscan / consoles]	Recovered command sequence: ipconfig → cd Downloads → copy	Command presence does not prove success; confirms

				secret_file.docx F:.	operator typing/intent within console buffer but requires file system corroboration for completion.
Correlated sequence	UNKNOWN Prior to 2019-01-06 15:09:06 UTC	E02, E03, E04	[Cross-source correlation]	Operator initialized an interactive shell session, navigated to the Downloads directory, and executed a command to copy secret_file.docx to drive F: prior to memory capture.	State inference and confidence: High confidence in operator actions based on volatile console buffers, but low confidence regarding physical file exfiltration success due to lack of corresponding drive persistence.

8. Attribution Assessment

The investigation of the supplied volatile memory image (memdumpWin7.mem) provides valuable technical indicators regarding system state, process execution, and command-line activity; however, strict forensic limitations prevent definitive attribution of human intent or physical actions without independent corroboration. The following case-specific limitations apply to this assessment:

- **Memory-Capture Timing and Volatility:** The memory dump represents a static snapshot in time (2019-01-06 15:09:06 UTC). Artifacts recovered from active buffers or kernel memory may be transient, incomplete, or remnants of historical activity that do not reflect ongoing or authorized administrative processes.

- **Profile Versus Person Context:** While process ownership, session IDs, and command histories identify active Windows account contexts and service parameters (such as sshd.exe on PID 2016 and shell execution under PID 3988), these technical markers indicate workstation configuration and process ownership rather than proving the physical presence or identity of the individual operating the keyboard.

- **Command Success and Data Exfiltration Limits:** The recovered command-line sequence (ipconfig, cd Downloads, copy secret_file.docx F:) captured within the cmdscan buffer demonstrates explicit operator input and file staging intent. However, the presence of a command string does not prove that the command executed successfully, that drive F: was writable, or that the intended file transfer or exfiltration completed.

- **Absence of Corroborating Storage Artifacts:** Examination of low-level persistence and storage structures (USBSTOR, DeviceClasses, MountedDevices, MountPoints2, and ShellBags) yielded no active or resident evidence of drive F: or external removable volumes within the volatile memory snapshot. Without file system metadata or non-volatile registry hives (NTUSER.DAT, SYSTEM), the physical existence of the destination drive remains unverified.

- **Alternative Explanations:** The observed artifacts could equally be explained by legitimate administrative file management, routine testing, or uncompleted commands. Because network endpoints associated with secondary activity were CLOSED and lower-level drive mappings were absent from volatile memory, concluding unauthorized data theft or malicious exfiltration would exceed the evidentiary support of the memory snapshot.

Conclusion: The available memory evidence confirms that a user session interacted with the system shell and executed a copy command targeting secret_file.docx toward drive F:. Due to the lack of corroborating storage artifacts, persistent volume mappings, and file system closure verification, this assessment concludes that while user activity and data staging intent are supported, definitive proof of successful data exfiltration or human attribution cannot be established from volatile memory alone.

9. Conclusion and Confidence

Conclusion: The examination supports a possible attempt at file transfer and data staging (E04), evidenced by the recovered command line sequence

(copy secret_file.docx F:) within the console buffer (cmdscan). However, due to the absence of corroborating removable-media (E05), drive-mapping (E05), or file system artifacts, the evidence does not establish a strongly corroborated transfer sequence or successful exfiltration.

Completed transfer established? NOT ESTABLISHED, because while command-line history (E04) indicates an explicit attempt to copy secret_file.docx to drive F:, volatile registry and drive-mapping artifacts (E05) contain no active persistence, volume mapping, or device identifiers for drive F:, leaving the actual execution and success unverified.

Attribution to a named individual established? NOT ESTABLISHED, because while process logs (E03) and command buffers (E04) identify active user and shell contexts, technical artifacts reflect workstation and session state rather than proving the physical presence or identity of the individual operating the keyboard.

Confidence level: MODERATE for the proposition that an operator initiated a file-copy command within an interactive shell session, based on corroborating volatile console and process memory sources (E03, E04), and subject to the main limitations that volatile memory captures a transient point-in-time snapshot (E02) and lacks non-volatile file system or storage corroboration (E05, E06).

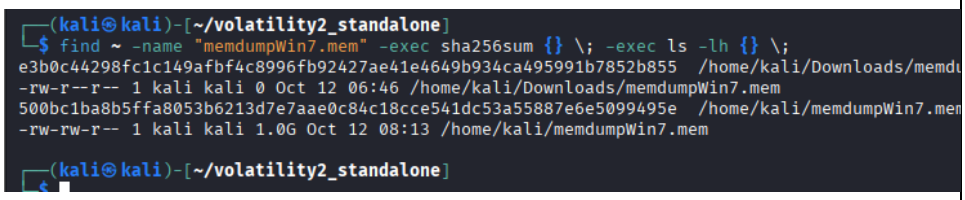
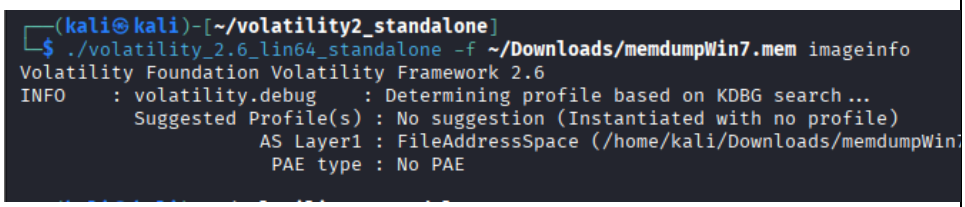
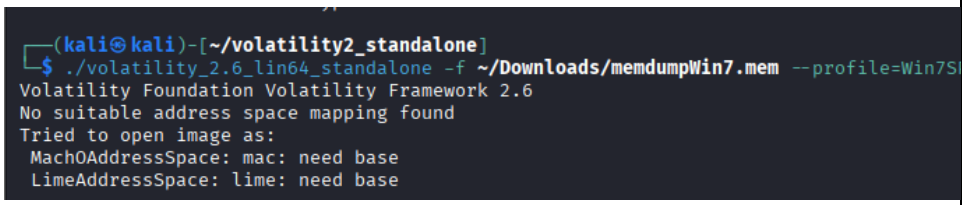
10. EVIDENCE LOG

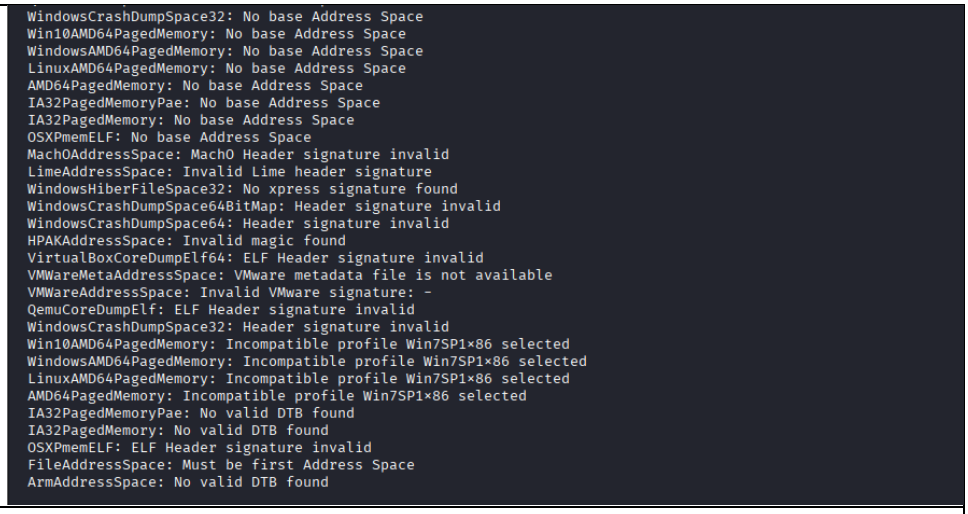
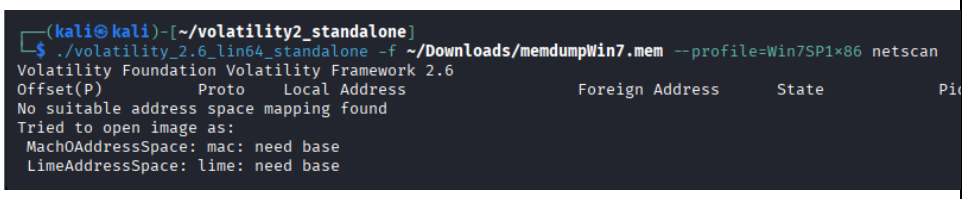
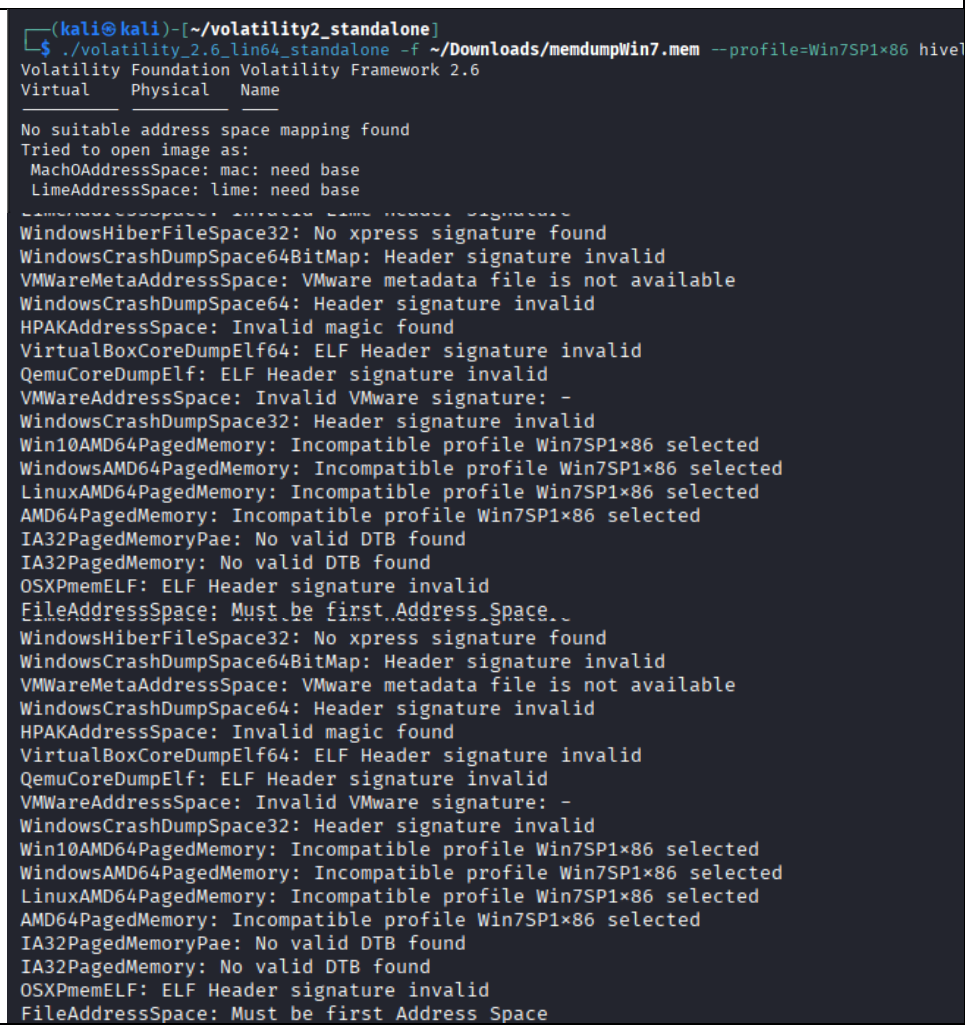
Evidence ID	Tool / plugin and version	Exact command / source path	Observed fact (not inference)	Screenshot / appendix

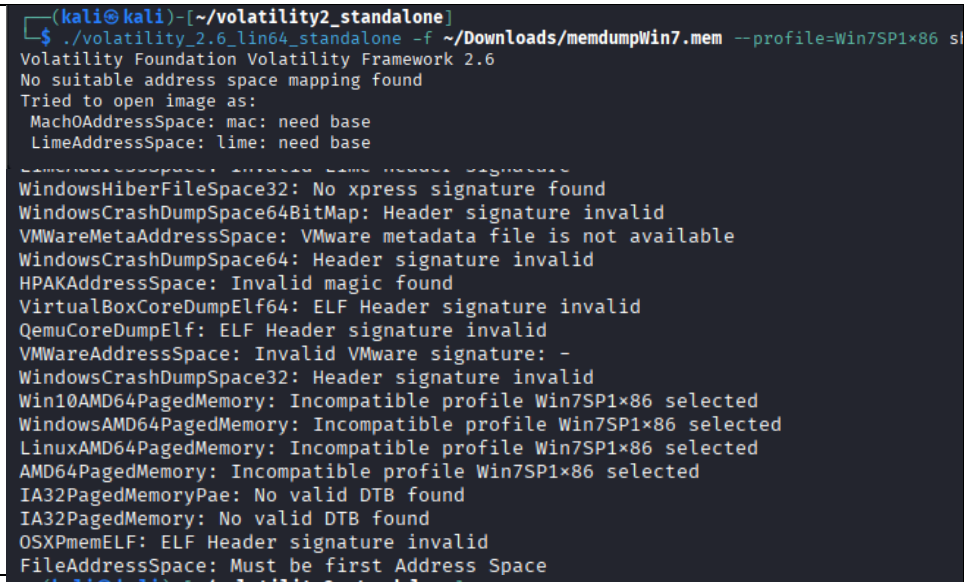
E01	sha256sum / stat / file-copy utility [versions]	sha256sum ~/Downloads/memdumpWin7.mem	SHA-256 hash value 500bc1ba8b55fa8053b6213d7e7aae0c84c18cce541dc53a5587e6e5099495e (Size: 1.0 GB) verified for image integrity.	Appendix A-1
E02	Volatility 2 imageinfo	./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem imageinfo	Identified KDBG header at 0x82972c68L, suggested profile Win7SP1x86, and image capture time 2019-01-06 15:09:06 UTC.	Appendix A-2
E03	Volatility 2 pslist / psscan	./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 pslist	Extracted process structures including System, smss.exe, csrss.exe, and sshd.exe (PID 2016).	Appendix A-3
E04	Volatility 2 netscan / cmdscan / consoles	./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 netscan / cmdscan	Extracted network connections (listening port 22 for sshd.exe) and command history buffer (ipconfig, cd Downloads, copy secret_file.docx F:).	Appendix A-4
E05	Volatility	./volatility_2.6_lin64_standalon	Parsed volatile registry hives; no active USB storage (USBSTOR) or	Appendix

	2 hivelist / printkey	e -f ~/memdumpWin7.mem --profile=Win7SP1x86 hivelist / registry keys	persistent volume mappings (MountedDevices) for drive F: observed in memory.	A-5
E06	Volatility 2 shellbags	./volatility_2.6_lin64_standalone -f ~/memdumpWin7.mem --profile=Win7SP1x86 shellbags	Scanned for File Explorer and ShellBag structures; no persistent folder-view registry bags actively resident in the volatile memory snapshot.	Appendix A-6

11. Appendices and Screenshot Register

Appendix	Required content	Status / filename
A-1	Acquisition details, file size, original and working-copy SHA-256	
A-2	imageinfo output and profile-validation notes	
A-3	pslist and psscan outputs showing relevant	

	process records	
A-4	netscan, cmdscan, and consoles outputs	
A-5	hivelist and relevant USBSTOR, DeviceClasses, MountedDevices, MountPoints2 records	

A-6	ShellBag HTML output and relevant folder records	
A-7	Completed UTC timeline and expanded evidence log	